GenCC Recommendation for NOT Loss of Function (NOT LOF) mechanism curation - Version 1.0

The Gene Curation Coalition proposes a semi-quantitative metric to determine the mechanism of disease for gene-disease relationships. The framework is intended to be used for human gene-disease relationships that have scored at Moderate or higher using the ClinGen gene curation framework (Strande et al 2017, see here for the most recent gene curation SOP). Levels should be capped at "Suspected" if applying to genes with Limited or below evidence. Caution should be exercised if adapting to non-human genetic diseases as this framework has not been tested using non-human phenotypes.

This framework is for evaluating genes for an assertion that loss of function (LOF) is NOT the disease mechanism. Of note, curators can assert that LOF is NOT the mechanism of disease when the true mechanism is still unclear. Gene-disease relationships can be evaluated for monoallelic and biallelic Not LOF.

**While this is a semi-quantitative framework, please note that points are not statistical in nature, nor do they correspond to Bayesian points in the sequence variant classification frameworks.

Table 1. Matrix for Biallelic NOT Loss of Function Framework

Variant Type	Evidence	Points
pLOF Variant/Gene deletions	pLOF variants or gene deletion identified in a healthy individual (with correct zygosity)	Unaffected - 0.2 pts Different phenotype from one under evaluation - 0.1 pts
Experimental Evidence	KO model organism does NOT have a phenotype	0.3 pts (0-0.5), depending on phenotype being recapitulated in a model and given the authors investigated the phenotype under evaluation
	Functional experiment demonstrates LOF is not a mechanism (gene-level or multiple variants)	0.1 pts (0.1-0.3) depending on quality, confidence that you are evaluating LOF, and number of experiments)

Lack of Consistency in population databases	Presence of high frequency homozygous variants of the type above, considering the disease severity, age of onset and known presence of affected individuals in gnomAD from biobanks, etc	0.3 pts (0.1-0.3) Score can vary depending on the number of variants and type of condition (onset, penetrance)

Footnotes:
Abbreviations: LOF, Loss of Function; pLOF, predicted LOF; KO, knock out

Table 2. Matrix for Monoallelic NOT Loss of Function Framework

Variant Type	Evidence	Points
pLOF Variant/Gene deletions	pLOF variants or gene deletion identified in a healthy individual (with correct zygosity)	Unaffected - 0.2 pts Different phenotype from one under evaluation - 0.1 pts
Experimental Evidence	KO model organism does NOT have a phenotype	Het KO 0.3 points Hom KO 0.15 points Also consider the phenotype being recapitulated in a model and given the authors investigated the phenotype under evaluation
	Functional experiment demonstrates LOF is not a mechanism	0.1 pts (0.1-0.3) depending on quality, confidence that you are evaluating LOF, and number of experiments)
Lack of Consistency in population databases	Lack of constraint for LOF in gnomAD considering the disease severity, age of onset, and proposed inheritance of the disease	0.15 pts

Table 3. Summary Matrix for NOT LOF Frameworks

Points from Tables 1 or 2 should be summed and the equivalent confidence term applied to the NOT LOF mechanism

Evidence Points	Confidence Term	Mechanism
.99 or higher	Established	Not Loss of Function
0.90-0.98	Likely	
0.5-0.89	Suspected	
0.49-0	Uncertain	